Mohamed Osama Zahran

Software Engineer

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years of production delivery — four parallel enterprise systems at RICOH Europe (technical design through implementation, testing, deployment and stabilization), a live multi-portal government healthcare application, and e-commerce products in production with real users. Works across the full stack: React 19 + Next.js + TypeScript frontends (reusable component libraries, modular state management, WCAG + Arabic RTL/i18n, performance and bundle optimization) and the APIs behind them (ASP.NET Core, NestJS, FastAPI, REST contracts, JWT auth, SQL and NoSQL data, SignalR real-time), plus AI systems in production (LangGraph multi-agent pipelines, LLM tool-calling + SSE streaming, HuggingFace transformers). 390+ automated Vitest tests gating CI/CD pipelines; reviews code and translates complex workflows into clean, fast, reliable software within cross-functional product teams.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, SQL

Frontend (React ecosystem): React 19, Next.js 15/16, TypeScript, JavaScript (ES6+), HTML5/CSS3, e-commerce storefronts & dashboard applications, reusable component libraries & design systems (Shadcn / Radix UI), hooks & modern patterns, state management (Redux Toolkit, Zustand, TanStack Query/Form), REST API integration with secure JWT auth & RBAC, responsive & accessible interfaces (WCAG), React Router, modular & scalable front-end architecture, performance & bundle optimization, RTL + next-intl i18n

Backend & APIs (working knowledge): NestJS, ASP.NET Core, EF Core, MongoDB, SQL Server, PostgreSQL, Prisma, REST API contracts, SignalR/WebSockets

AI & Integrations: LangGraph multi-agent pipelines, LLM tool-calling + SSE streaming, HuggingFace Transformers (Llama 3, Qwen 2.5, Whisper, DistilBERT), Google Gemini, Redis

Styling & UI: Tailwind CSS v4, Shadcn / Radix UI design system, Ant Design, Framer Motion, GSAP, Lenis, RTL + next-intl i18n, WCAG accessibility

Tools & Practices: Git (pull requests, code reviews), GitHub Actions CI/CD with quality gates, Vitest (390+ tests), xUnit, Zod, Swagger/OpenAPI, Docker & docker-compose, Vercel + containerized cloud deployment, Lighthouse & performance profiling, technical-debt identification & sustainable remediation, code-standards ownership

PROFESSIONAL EXPERIENCE

RICOH Europe — Maadi, Cairo *Nov 2025 – Present*

Full Stack Engineer

- Own front-end delivery across 4 production enterprise systems in parallel (CORELIA program, cross-regional European teams): technical design through implementation, testing, deployment and stabilization of responsive, accessible interfaces; set component patterns, code standards and review practices for the front-end surface.
- Collaborate daily with multicultural, cross-functional teams in a corporate European environment.

NedSwiss — Remote, Switzerland Oct 2023 – Nov 2025

Full Stack Engineer

- Built a multilingual admin/staff CRM dashboard (Next.js 15, TanStack Table/Query, Recharts, react-quill) with role-based navigation, rich data tables, chart analytics, and real-time cache invalidation.
- Developed the company marketing website with advanced scroll animations (GSAP, Lenis), next-intl i18n routing, Zod-validated lead capture, and Turbopack bundle optimization.

GDSC – Shorouk Academy — Shorouk City *Aug 2023 – Jun 2025*

Frontend Lead

- Led the frontend track of the Google Developer Student Club: designed and ran hands-on workshops in React, Next.js, and TypeScript, and coached developers through modern front-end architecture patterns.

- Mentored developers in React, TypeScript, state management, and modular UI architecture; authored workshop material reused across cohorts.

PROJECTS

RICOH Production Work — CORELIA (Nov 2025 – Present)

- **Tailgating Detection System:** real-time anomaly dashboard with live MJPEG streaming, license-plate OCR, role-based access, and audit trails (React 18, Redux Toolkit, TanStack Query, Shadcn UI).
- **3D Building & Space Viewer:** multi-mode 3D viewer (Dollhouse / Panorama / Split) with room-aware navigation, animated overlays, and progressive panorama loading (React 19, React Three Fiber, GSAP, Framer Motion).
- **AGL Invoice Management:** full invoice lifecycle with React Hook Form + Zod integrated with CV team and AI Team for the full pipeline.
- **Qalam Vision:** Arabic/multilingual NLP keyword-extraction interface integrating Label Studio.

OZ Uniform Store — Full-Stack Client Product *(Jul 2026 – Aug 2026)*

Next.js 16, React 19, Tailwind CSS v4, TanStack Form + Zod 4, .NET 10 Web API, EF Core 10, SQL Server, Hangfire

- Arabic-first RTL e-commerce storefront for a school-uniform retailer: school/grade catalog, cart, cash-on-delivery checkout, Bosta courier shipping, and per-order token tracking URLs.
- Built the .NET 10 backend: JWT auth, Hangfire background jobs (auto-cancel orders, email), FluentValidation, Bosta shipping integration, and audit logging; deployed to production.

Medical System — Gov Healthcare Insurance Portal (Aug 2026)

Next.js 16, React 19, TypeScript, TanStack Query, React Hook Form + Zod, next-intl (AR/EN RTL), Recharts, Shadcn / Radix UI, Tailwind CSS, Biome, Vitest

- Co-built a production government healthcare portal (citizen, doctor, and admin portals, live on Vercel) with a 2-developer team, contributing the majority of commits (40 of 59): the full admin verification workflow across 4 phases (application review queue, card lifecycle management, category configuration, verification workbench).
- Built the doctor clinical workspace (point-of-care desk with patient chart and visit management) and the citizen insurance experience (application, tracking history, card-issuance stepper) on OpenAPI-snapshot API contracts with role-based auth.

ARME Egypt (Arab River Medical) — B2B Medical Equipment Portal (May 2026 – Jun 2026)

Next.js 16, React 19, TypeScript, Tailwind CSS v4, next-intl (EN/AR RTL), Docker

- Bilingual English/Arabic (RTL) B2B portal for a surgical-instruments and medical-furniture supplier: public marketing site with context-aware lead capture that geo-routes Egyptian visitors to a catalog-download flow and international visitors to an agency-inquiry flow.
- Secure admin dashboard for catalogs, lead review, and analytics; JWT session handling, honeypot anti-spam, XSS-safe JSON-LD, WCAG AA contrast, and a dynamic sitemap; live in production.

Wasla CRM — Multi-Tenant SaaS (Graduation Project) (Nov 2025 – Jun 2026)

Next.js 15, ASP.NET Core 8, FastAPI, SignalR, Hangfire, Stripe, Paymob, HuggingFace

- Full CRM platform spanning frontend, .NET 8 backend, and two AI microservices: multi-tenant JWT auth with per-user permissions, SignalR real-time notifications, Hangfire scheduled jobs
- **AI layer**: Llama-3.3/Qwen tool-calling backend, and **toxicity moderation** (DistilBERT / DeHateBERT) with sub-100ms latency on high-confidence predictions.

Nabdh Platform — Home Nursing Marketplace Backend (Jun 2026 – Present)

NestJS 10, TypeScript, MongoDB 7 (2dsphere geospatial), WebSocket, MinIO S3, Docker

- Engineered the modular-monolith backend for an Egyptian home-nursing marketplace: auth, booking, payment, chat, and notification modules serving patient and nurse Flutter apps as well as a React admin panel.
- Geospatial matching with MongoDB 2dsphere indexes, real-time WebSocket chat, and S3-compatible media storage via MinIO.

CyrusLearn — Interactive Learning Platform (WE Innovative Hackathon) *(Mar 2026 – Apr 2026)*

React 19, Vite, TypeScript, React Three Fiber, Redux Toolkit, Shadcn UI

- Feature-driven e-learning platform with 3D interactive labs, structured learning paths, a quiz engine, collaborative rooms, Google OAuth, subscription flow, and an admin dashboard.

MAC-Egypt — UNIDO/NOU/EED Assessment & Monitoring Platform (Oct 2026 – Present)

Next.js 16, React 19, TypeScript, Tailwind CSS v4, shadcn/ui, next-intl (Arabic-first RTL)

- Freelance build for a UNIDO/NOU/EED program: workshop registry with duplicate detection, a 13-section field survey wizard with GPS and photo evidence, machine-scored selection runs, training and equipment-delivery records, and role-shaped dashboards.
- Bilingual Arabic-first RTL/English UX on a schema-driven survey engine; 409 passing tests gating a GitHub Actions deploy pipeline shipping to production.

Meapal LogBook — Team Activity Reporting for RICOH/Corelia (Sep 2026)

React 19, Vite, TypeScript, Tailwind CSS, shadcn/ui

- Daily activity log for RICOH/Corelia teams: calendar-driven pay-period tracking against hour targets, work streaks with rest-day-aware logic, weekly analytics, reminders, and recurring-entry templates.
- Three role surfaces (employee, project manager, admin) with analytics and administration views.

agentic-shop — Agentic AI Shopping System (Aug 2026)

Python, FastAPI, LangGraph, Pydantic v2, Next.js 16, React 19, SSE

- Built a multi-agent shopping pipeline (LangGraph) that turns a one-sentence goal into ranked products, a conversational answer, and a validated UI plan rendered client-side over SSE streaming.
- Deterministic product scorer with Pydantic v2 contracts; fully offline mock mode with 390 passing tests (224 backend + 166 frontend); real-LLM swap-in ready.

Attendance API — GPS-Verified Workforce Service (Aug 2026)

NestJS 11, TypeScript, MongoDB (Mongoose 9), JWT + Passport, Swagger/OpenAPI, class-validator

- Built a NestJS service for GPS-verified employee attendance: per-workplace geofencing, timezone-aware working days, and a calendar history that classifies each day as present, incomplete, absent, weekend, or upcoming.
- JWT authentication with bcrypt hashing, rate-limited endpoints, OpenAPI/Swagger documentation, and a published Postman collection.

Gym Management API (Aug 2026)

ASP.NET Core 8, EF Core, SQL Server, xUnit, Swagger/OpenAPI

- Single-tenant gym subscription API with cryptographic QR check-in: atomic session deduction under concurrency (SQL applocks + conditional updates), preventing negative balances.
- Time-based and per-sale session-credit subscriptions, RFC 7807 errors with machine-readable reason codes, structured logging, health checks, and an xUnit test suite.

NED Swiss — Digital Agency Website (Oct 2023 – Nov 2025)

Next.js 15, React 19, TypeScript, Tailwind CSS v4, next-intl (EN/DE/FR), GSAP, Framer Motion, Lenis

- Trilingual (English/German/French) marketing website for a Swiss digital agency: locale routing, GSAP scroll animations, Lenis smooth scrolling, magnetic hover micro-interactions, and per-locale SEO metadata; in production at ned-swiss.ch.

NED Swiss — Admin/Staff CRM Dashboard (Oct 2023 – Nov 2025)

Next.js 15, TypeScript, TanStack Table + TanStack Query, Recharts, react-quill

- Multilingual admin/staff CRM dashboard: rich data tables with server-side mutations, chart analytics, rich-text editing, and real-time data sync across Arabic and English locales.

AI-Powered Translation Platform — Freelance (2024)

Next.js, TypeScript, Tailwind CSS, next-intl, Paymob

- Multilingual translation product (Arabic/English/French) with end-to-end Paymob billing, a daily-quota credit system, admin transaction monitoring, and a blog module.

EDUCATION

BSc in Computer Science — Shorouk Academy, Cairo *GPA: 3.8/4.0*